Structural Changes in the Conditional Volatility Process of Stock Market Returns

Josip Arnerić
Department of Quantitative Methods
University of Split, Faculty of Economics
Matice hrvatske 31, 21000 Split, Croatia
e-mail: jarneric@efst.hr

Abstract—**the most successful models in modeling time varying volatility are GARCH type models. However, when financial returns exhibit sudden jumps that are due to structural breaks the GARCH models show high volatility persistence, i.e. integrated behavior of the conditional variance. In such situations models in which the parameters are allowed to change over time are more appropriate. This paper compares different GARCH models in terms of their ability to describe structural changes in returns caused by financial crisis at Croatian stock market. Fat-tailed conditional distribution of innovations is assumed. Moreover, state dependent degrees of freedom are assumed to model possible time varying kurtosis. The empirical analysis demonstrates that Markov regime switching GARCH model resolves the problem of excessive persistence and outperforms uni-regime GARCH models in forecasting volatility when sudden switching occurs in response to financial crisis. Regime switching GARCH models allow different speeds of mean reversion of innovation process on different levels of variance in different time periods. The data set used in the study consists of returns of the CROBEX index daily closing prices obtained from Zagreb Stock Exchange.**

Keywords - risk management, structural changes, Markov switching process, conditional volatility, transition probabilities

I. INTRODUCTION

Portfolio managers, option traders and market makers are all interested in volatility forecasting in order to get higher profits or less risky positions. In the last few decades there has been enormous interest in forecasting of returns fluctuations. Based on the fact that volatility is time varying in high frequency data and that periods of high volatility tend to cluster, the most successful and popular models in modeling time varying volatility are GARCH type models. The first autoregressive conditional heteroscedasticity model (ARCH) was proposed by Engle (1982) who won a Nobel Prize in 2003 for his contribution to modeling volatility. The model was extended by Bollerslev (1986) by its generalized version (GARCH). The current conditional volatility in GARCH(p,q) model is defined as a linear function of squared innovations in previous p periods and conditional variances in previous q periods. However, GARCH(1,1) model usually indicate high persistence in the conditional variance, which may originate from structural changes in the variance process. Hence the estimates of a GARCH model suffer from a substantial upward bias in the persistence parameters. Therefore, models in which the parameters are allowed to change over time may be more appropriate for modeling volatility. The main feature of regime switching model is the possibility for some or all the parameters of the model to switch across different regimes according to a Markov process, which is governed by a state variable S_t.

Application of Markov regime switching models in the volatility modeling of financial time series has significantly increased in recent years, primarily due to the simplicity of the interpretation of nonlinear behavior of returns, which originates from structural changes in the variance process. Regime switching GARCH models allow different speeds of mean reversion of innovation process on different levels of variance in different time periods. Hence, in this paper Markov regime switching GARCH model (MRS-GARCH), where the parameters are allowed to switch between low and high volatility regime, is analyzed to describe structural changes in returns caused by financial crisis at Croatian stock market.

The structure of the paper is organized as follows. In the second section integrated behaviour of the conditional variance is described. In the third section Markov switching GARCH model is presented, while the fourth section summarizes the estimation results of the research. Last section provides conclusion of the empirical findings.

II. INTEGRATED BEHAVIOUR OF THE CONDITIONAL VARIANCE

The most widespread approach to modeling volatility consists of the GARCH model of Bollerslev (1986) and its numerous extensions that can account for the volatility clustering and excess kurtosis found in financial time series. The accumulated evidence from empirical research suggests that the volatility of financial markets can be appropriately captured by standard GARCH(1,1) model:

$$\begin{aligned} & r_t = \mu_t + \varepsilon_t;\ \varepsilon_t = u_t \cdot \sqrt{\sigma_t^2} \\ & u_t \sim i.i.d.(0,1) \\ & \sigma_t^2 = \alpha_0 + \alpha_1 \cdot \varepsilon_{t-1}^2 + \beta_1 \cdot \sigma_{t-1}^2, \end{aligned} \tag{1}$$

where μ_t is the conditional mean of return process $\{r_t\}$, while $\{\varepsilon_t\}$ is the innovation process with its multiplicative

978-1-4244-8618-2/10/$26.00 © 2010 IEEE

Authorized licensed use limited to: University of Sydney. Downloaded on September 16,2026 at 12:57:59 UTC from IEEE Xplore. Restrictions apply.

structure of identically and independently distributed random variables u_t. The last equation in (1) is conditional variance equation with GARCH(1,1) specification which means that variance of returns is conditioned on the information set I_{t-1} consisting of all relevant previous information up to period *t-1*. By substitution $v_t = \varepsilon_t^2 - \sigma_t^2$ the GARCH(1,1) model can be rewritten as an ARMA(1,1) model for ε_t^2 as:

$$\varepsilon_t^2 = \alpha_0 + (\alpha_1 + \beta_1) \cdot \varepsilon_{t-1}^2 + v_t - \beta_1 \cdot v_{t-1} \quad (2)$$

According to ARMA(1,1) representation in (2) it follows that GARCH(1,1) model is covariance-stationary if and only if $\alpha_1 + \beta_1 < 1$. In particular, GARCH(1,1) model usually indicates high persistence in the conditional variance, i.e. integrated behaviour of the conditional variance when $\alpha_1 + \beta_1 = 1$ (IGARCH). The reason for the excessive GARCH forecasts in volatile periods may be the well known high persistence of individual shocks in those forecasts. Lamoureux and Lastrapes (1990), among others, show that this persistence may originate from structural changes in the variance process. They demonstrate that shifts in the unconditional variance lead to biased estimates of the GARCH parameters suggesting high persistence. High volatility persistence means that a long time period is needed for shocks in volatility to die out (mean reversion period). These findings clearly indicate a potential source of misspecification, to the extent that the structural form of the conditional mean and variance is relatively inflexible and held fixed throughout the entire sample period. Wong and Li (2001) demonstrate that existence of shifts in the variance process over time can induce volatility persistence. Therefore, models in which the parameters are allowed to change over time may be more appropriate for modeling volatility. A popular approach to endogenize changes in the data generating process is the Markov regime switching model. Hamilton (1989) introduces this model to describe the U.S. business cycle, which is characterized by periodic shifts from recessions to expansions and vice versa. In our context of stock market volatility, a Markov process can be used to govern the switches between regimes with different variances. In practical applications, the structural changes are rarely observed in advance and state variable cannot be determined "a priori". If such changes can be predetermined then the model becomes a model with dummy variable.

In addition the kurtosis of ε_t is always larger than the normal value of 3 even if Gaussian distribution is assumed. This excess kurtosis implies that conditional distribution has fatter tails than the normal distribution, which means that large observations occur much more often than one might expect for a normally distributed variable. Therefore, Markov switching models can be used for modeling possible time varying kurtosis when degrees of freedom are state dependent variable assuming Student t-distribution.

III. MARKOV SWITCHING GARCH MODEL

In order to account for the presence of structural changes in the variance process, Hamilton and Susmel (1994) were first who applied the idea of regime switching parameters by combining Markov switching model with ARCH specification. Markov switching ARCH model was designed to capture regime changes in volatility with unobservable state variable S_t following the first order Markov Chain with constant transition probabilities. The logic behind this kind of modeling is having a mixture of distributions with different characteristics.

Hamilton and Susmel (1994) used an ARCH specification instead of a GARCH to avoid infinite path dependence problem. However, some proposed MRS-GARCH models must be taken with caution because the stationarity conditions are not well known, or whose dynamic properties are not sufficiently tested. It should also be noted that most regime switching models appear certain difficulties in parameters estimation. Therefore, several models with certain modifications and restrictions are proposed. The most general form of MRS-GARCH(1,1) model can be written as:

$$\begin{aligned} r_t &= \mu_{S_t} + \varepsilon_t \\ \varepsilon_t &= u_t \cdot \sigma_{t,S_t} \\ u_t &\sim i.i.d.(0,1) \\ \sigma_{t,S_t}^2 &= \alpha_{0,S_t} + \alpha_{1,S_t} \cdot \varepsilon_{t-1}^2 + \beta_{1,S_t} \cdot \sigma_{t-1}^2, \end{aligned} \quad (3)$$

where the unobserved state variable S_t is assumed to evolve according to the first-order Markov chain, with transition probability:

$$\Pr(S_t | S_{t-1},\ S_{t-2}, ..., S_1, I_{t-1}) = \Pr(S_t = j | S_{t-1} = i) \quad (4)$$

that indicates the probability of switching from state *i* at moment *t-1* into state *j* at moment *t*. When only two regimes are considered $S_t = \{1, 2\}$, i.e. low and high volatility regimes, these probabilities can be grouped together into probability transition matrix:

$$P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} = \begin{bmatrix} p & (1-p) \\ (1-q) & q \end{bmatrix} \quad (5)$$

Probabilities p_{11} and p_{12} are complementary, just as the probabilities p_{21} and p_{22}. Based on transition probabilities $\Pr(S_t = j | S_{t-1} = i)$ the conditional probabilities

Authorized licensed use limited to: University of Sydney. Downloaded on September 16,2026 at 12:57:59 UTC from IEEE Xplore. Restrictions apply.

$\Pr(S_t = j | I_{t-1})$ can be generated to define the likelihood function when normal distribution is assumed:

$$\ln L = \sum_{t=1}^{T} \ln \left\{ \left(\sum_{j=1}^{2} \frac{1}{\sqrt{2\pi\sigma_{S_t}^2}} \exp\left[-\frac{(r_t - \mu_{S_t})^2}{2 \cdot \sigma_{S_t}^2} \right] \right) \cdot \Pr(S_t = j | I_{t-1}) \right\} \quad (6)$$

Hence, in order to maximize the likelihood function the stochastic behavior of discrete state variable must be assumed. In the brackets of expression (6) probability density function for each observation is presented as a weighted sum of the conditional distribution functions for both regimes. However, Student t-distribution or GED distribution can also be assumed. Since the procedure of maximizing the likelihood function for parameters estimation is iterative in each new iteration the conditional probabilities can be updated using Kim's filter (smoothing algorithm). For the initial value of the variance by recursive substitution in the conditional variance equation of expression (3) it can be obtained:

$$\sigma_t^2 = \sum_{i=0}^{t-1} \left[\alpha_{0,S_{t-i}} + \alpha_{1,S_{t-i}} \cdot \varepsilon_{t-1-i}^2 \right] \prod_{j=0}^{i-1} \beta_{1,S_{t-j}} + \sigma_0^2 \prod_{i=0}^{t-1} \beta_{1,S_{t-i}} \quad (7)$$

Expression (7) shows that depends on the entire history of regimes, i.e. there is an infinite path dependence problem, which means that conditional variance at moment *t* depends on the conditional variance at moment *t-1* and state variable at moment *t* while the conditional variance at moment *t-1* depends on the conditional variance at moment *t-2* and state variable at moment *t-1*, and so on. Then, for two regime states, the evaluation of the likelihood function requires the integration over all possible regime paths for each observation, which makes the estimation infeasible in practice. In order to solve the problem of path dependence Gray (1996) proposed that conditional variance in variance equation is given by information at moment *t-2*, i.e. he suggests to substitute σ_{t-1}^2 with conditional expected value $E_{t-2}(\sigma_{t-1}^2)$. Klaassen (2002) introduced modification of Gray's MRS-GARCH(1,1) model by using conditional variance of ε_{t-1} given the information *t-1*, i.e. he suggests to substitute σ_{t-1}^2 with conditional expected value $E_{t-1}(\sigma_{t-1}^2 | S_t)$. This makes Klaassen's specification more and therefore, in this paper, Klaassen's MRS-GARCH(1,1) model is compared to standard GARCH(1,1) model.

IV. Empirical Research

The data set analyzed in this paper is the CROBEX stock market daily closing price index. The sample period is from January 03, 2005 to April 30, 2009 for a total of 1105 observations, all obtained from Zagreb Stock Exchange. Both the conditional mean and the conditional variance equations are estimated jointly by maximizing the log-likelihood function which is computed as the natural logarithm of the product of the conditional densities of the innovations, which are assumed to follow Student-t distribution. When estimating MRS-GARCH(1,1) model each regime takes different degrees of freedom of a Student t-distribution. Parameters and are also allowed to switch between two regimes. Maximum likelihood estimates are obtained with Broyden, Fletcher, Goldfarb and Shanno (BFGS) numerical quasi-Newton optimization algorithm in the Time Series Modelling package (TSM 4.29).

TABLE I. GARCH Models Estimation With Diagnostic Tests

Parameters*	GARCH(1,1) t-distribution	MRS-GARCH(1,1) t-distribution
ϕ_0	0.00079**	0.00076**
α_0	0.00003*	0.000025*
α_1	0.1438**	0.2176**
β_1	0.8469**	0.6298*
df	4.6	4.93**
$\lim_{t\to\infty} \sigma_t$	3.22%	1.81%
p	-	0.98165**
q	-	0.95904**
$\alpha_{1,2}$	-	0.0258**
$\beta_{1,2}$	-	0.9553**
df_2	-	4.70**
k	4	10
$\ln L^*$	3358.9	3361.3
AIC	-6709.8	-6702.6
$SBIC$	-6689.8	-6652.5
$Q(10)$	5.4478	7.7996
$Q^2(10)$	5.9526	9.7704
$LM(10)$	5.8366	9.6348

a. The significance at 10% level is marked by (*), 5% level by (**) and 1% level by (***).

The estimation results (Table 1) show that high volatility persistence of 99,07% is associated with the standard uni-regime GARCH(1,1) model, which suggests that a long time is needed for shocks in volatility to die out (approximately 74 days). Even if the covariance-stationarity condition is satisfied the finite unconditional variance is overestimated (3,22%), due to the bias in the parameters. Specifically, in the standard GARCH(1,1) model 4 parameters were estimated, while degrees of freedom are pre-estimated using the method of moments (related to the kurtosis). In the case of MRS-GARCH(1,1) model degrees of freedom were

Authorized licensed use limited to: University of Sydney. Downloaded on September 16,2026 at 12:57:59 UTC from IEEE Xplore. Restrictions apply.

estimated jointly with other unknown parameters depending on the two regimes. Obviously, persistence volatility (0,8474) is lower in the regimes of low volatility compared to the persistence in the regime of high volatility (0,9811). This means that in the regime of low volatility mean reversion period is 4 days, while in the high volatility regime 36 days are needed for shocks in volatility to die out. Parameters in both estimated models are statistically significant.

Estimated transition probabilities indicate that the expected duration of the regime of low volatility of 54 days is less than the expected duration of the regime of high volatility (24 days). It is also important to point out that the variability of degrees of freedom of the Student t-distribution in MRS-GARCH(1,1) model suggests that degrees of freedom for a period of low volatility are greater in comparison to a period of high volatility. Specially, the estimated lower degrees of freedom in periods of increased volatility indicate an excess kurtosis in the crisis period, with fatter tails. It means that the occurrence of extremely low (negative) returns, as well as extremely high (positive) returns, is more frequent. Ljung-Box statistics, and Lagrange multiplier test are not statistically significant, suggesting that there is no ARCH effects or autocorrelation of standardized residuals (mean and variance equations are correctly specified). The value of AIC and SBIC information criteria is less in a standard GARCH (1,1) model compared with MRS-GARCH (1,1) model, because 10 parameters are estimated in regime switching model.

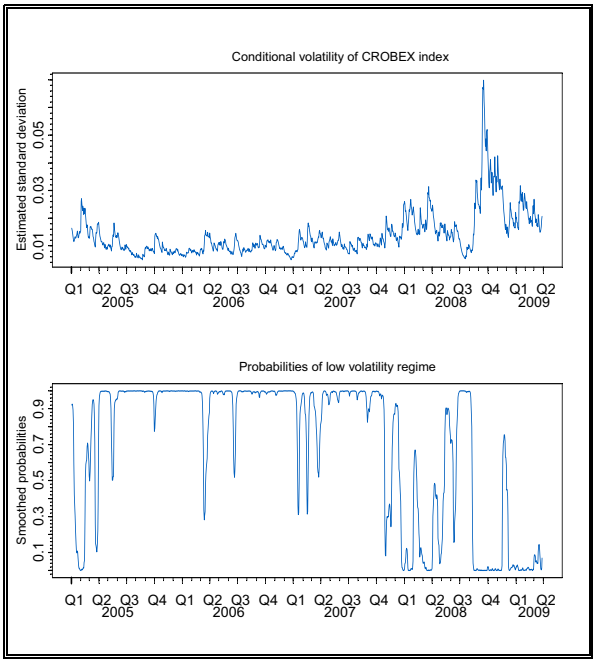


Figure 1. Conditional volatility of CROBEX index and smoothed probabilities of low volatility regime

Figure 1 shows smoothed probabilities that the return process is in a low volatility regime according to estimated MRS-GARCH(1,1) model. It is obvious that most probabilities in the period of financial crisis to the last day of observation are equal to zero, while the same probabilities in pre-crisis period are mostly equal to one.

V. CONCLUSION

Due to the effects of structural changes in the return series, caused by the financial crisis, sum of the parameters in standard GARCH(1,1) model indicates high volatility persistence. In such situations models in which the parameters are allowed to change over time are more appropriate, i.e. Markov switching GARCH model. Regime switching GARCH models allow different speeds of mean reversion of innovation process on different levels of variance in different time periods. Hence, in this paper Markov regime switching GARCH model (MRS-GARCH), where the parameters are allowed to switch between low and high volatility regime is analyzed to describe structural changes in returns caused by financial crisis at Croatian stock market. Estimation results show that volatility persistence is lower in the regime of low volatility compared to the persistence in the regime of high volatility. This means that in a regime of low volatility mean reversion period is 4 days, while in the high volatility regime 36 days are needed for shocks in volatility to die out.

Estimated transition probabilities indicate that the expected duration of the regime of low volatility of 54 days is less than the expected duration of the regime of high volatility (24 days). It is also important to point out that the variability of degrees of freedom of the Student t-distribution in MRS-GARCH(1,1) model suggests that degrees of freedom for a period of low volatility are greater in comparison to a period of high volatility. Specially, the estimated lower degrees of freedom in periods of increased volatility indicate that the occurrence of extremely low (negative) returns, as well as extremely high (positive) returns is more frequent.

The empirical analysis demonstrates that the Markov regime switching GARCH model resolves the problem of excessive persistence and outperforms uni-regime GARCH models in forecasting volatility when sudden switching occurs in response to financial crisis.

REFERENCES

[1] Beaver, W. H., McAnally, M. L., Stinson Christopher, H.: „The information content of earnins and prices: A simultaneous equations aproach“, Journal od Accounting and Economics 23 (1997.), pp. 53-81.

[2] Alexander, C., (2008), *Practical Financial Econometrics*, John Wiley & Sons, New York

[3] Bauwnes, L., Preminger, A., Romboust, J. V. K., (2006), Regime Switching GARCH Models, *CORE Discussion Paper*, No. 11, 1-23

[4] Brooks, C., (2002), *Introductory econometrics for finance*, Cambridge University Press, Cambridge

[5] Carnero, M. A., (2004), Persistence and Kurtosis in GARCH and Stohastic Volatility Models, *Journal of Financial Econometrics*, Oxford University Press, Vol. 2, No. 2, 319-342

[6] Engle, R., (1982), Autoregressive Conditional Heteroskedasticity with Estimates of the Variance of U.K. Inflation, *Econometrica*, Vol. 50, 987-1108

Authorized licensed use limited to: University of Sydney. Downloaded on September 16,2026 at 12:57:59 UTC from IEEE Xplore. Restrictions apply.

[7] Engle, R., (2001), The Use of ARCH/GARCH Models in Applied Econometrics, *Journal of Economic Perspectives*, Vol. 15, No. 4, 157-168

[8] Frances, P. H., Dijk, van D., (2003), *Nonlinear Time Series Models in Empirical Finance*, Cambridge University Press, Cambridge

[9] Hall, P., Yao Q., (2003), Inference in ARCH and GARCH Models with Heavy-Tailed Errors, *Econometrica*, Vol. 71, No. 1, 285-317

[10] Hamilton, J. D., Susmel, R., (1994), ARCH and Changes in Regime, *Journal of Econometrics,* Vol. 64, 307-333

[11] Hass, M., Mittnik, S., Paolella, M. S., (2004), A New Approach to Markov-Switching GARCH Models, *Journal of Financial Econometrics*, Vol. 2, No. 4, 493-530

[12] Hodrick, R. J., Prescott, E. C., (1997), Postwar U.S. Business Cycless: An Empirical Investigation, *Journal of Money, Credit and Bnaking*, Vol. 29, No. 1, 1-16

[13] Klaassen, F., (2002), Improving GARCH Volatility Forecasts with Regime-Switching GARCH, *Empirical Economics*, Vol. 27, No. 2, 363-394.

Authorized licensed use limited to: University of Sydney. Downloaded on September 16,2026 at 12:57:59 UTC from IEEE Xplore. Restrictions apply.